

Effects of Soil Moisture on Root Length Density in Winter Wheat

Root length density (RLD) is a key parameter for describing root distribution in the soil profile. Understanding how soil moisture regulates root growth is essential for improving water management strategies in semiarid regions.

In this study, we measured RLD at three growth stages under four irrigation treatments. Soil samples were collected at 10 cm depth intervals down to 100 cm using a soil auger.

The results showed that RLD decreased with increasing soil depth. The highest RLD values were observed in the topsoil layer.

Materials and Methods

The field experiment was conducted at the experimental farm of the university. The soil type was classified as a silt loam with a bulk density of 1.35 g cm⁻³. Four irrigation treatments were arranged in a randomized complete block design with three replications.

Root samples were washed over a 0.5 mm sieve and scanned with a root scanner. Root length was calculated using image analysis software.

Daily soil moisture was monitored using time domain reflectometry sensors installed at depths of 10, 30, 50, 70 and 90 cm.

Results and Discussion

Root length density varied significantly among irrigation treatments.

Deficit irrigation promoted deeper root growth, while full irrigation concentrated roots in the upper soil layers.

A positive correlation was found between soil moisture and RLD in the 0-30 cm layer, but the correlation became negative below 60 cm.

These findings suggest that moderate water stress may enhance root exploration of deep soil layers and improve drought resistance.